

DPT Exercises

Exercise sheet 10

Niki Kilbertus

Exercise 1 ([repeated from sheet 9:] mle for uniform distribution). Let $X_1, \dots, X_n$ be iid RRVs from a uniform distribution $\text{Unif}(0, \theta)$ with unknown parameter $\theta > 0$. The pdf of X_i is $p(x_i | \theta) = \frac{1}{\theta} \chi_{[0, \theta]}(x_i)$.

- (i) Write down the likelihood function $L(\theta | x)$ for an observed sample $x = (x_1, \dots, x_n)$. Be careful with the indicator functions.
- (ii) Find the maximum likelihood estimator (mle) $\hat{\theta}_{\text{mle}}$ for θ .
[Hint: The likelihood function may not be differentiable with respect to θ , but the maximum value can be found by reasoning about monotonicity and where it is non-zero.]
- (iii) Calculate the expected value $\mathbb{E}[\hat{\theta}_{\text{mle}}]$. Is $\hat{\theta}_{\text{mle}}$ an unbiased estimator for θ ?
[Hint: First find the cdf of $\hat{\theta}_{\text{mle}}$, then its pdf, and then compute the expectation.]

Solution:

- (i) The likelihood function for an observed sample $x = (x_1, \dots, x_n)$ is

$$L(\theta | x) = \prod_{i=1}^n p(x_i | \theta) = \prod_{i=1}^n \frac{1}{\theta} \chi_{[0, \theta]}(x_i) = \left(\frac{1}{\theta}\right)^n \prod_{i=1}^n \chi_{[0, \theta]}(x_i).$$

The product of indicator functions $\prod_{i=1}^n \chi_{[0, \theta]}(x_i)$ is 1 if and only if $0 \leq x_i \leq \theta$ for all $i \in [n]$. This is equivalent to $0 \leq \min\{x_1, \dots, x_n\} =: x_{\min}$ and $x_{\max} := \max\{x_1, \dots, x_n\} \leq \theta$. Since $x_i \geq 0$ for all x_i (they come from $\text{Unif}(0, \theta)$), we thus find $L(\theta | x) = \left(\frac{1}{\theta}\right)^n \chi_{[x_{\max}, \infty)}(\theta)$ when all $x_i \geq 0$ (and zero otherwise), or equivalently

$$L(\theta | x) = \begin{cases} (1/\theta)^n & \text{if } \theta \geq x_{\max} \text{ and all } x_i \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(We assume $x_{\max} > 0$ to rule out the case $\theta = 0$.)

- (ii) Since $L(\theta | x) = 0$ when $\theta < x_{\max}$ and it is monotonically decreasing in θ for $\theta \geq x_{\max}$ its maximum is at the smallest value θ for which the likelihood is non-zero. This is at $\theta = x_{\max}$. The mle is $\hat{\theta}_{\text{mle}} = X_{\max} = \max\{X_1, \dots, X_n\}$.

In the “bus-waiting-example,” this means that if you’ve been waiting for the bus for (4, 1, 7, 12, 5, 9, 10, 2) minutes on consecutive work days and you model bus arrival time as uniform, then the maximum likelihood estimate of the intervals in which the bus goes (maximum waiting time) is 12 minutes.

- (iii) To find $\mathbb{E}[\hat{\theta}_{\text{mle}}] = \mathbb{E}[X_{\max}]$, we first find the cdf of $X_{\max}$

$$F_{X_{\max}}(t) = P(X_{\max} \leq t) = P(\max\{X_1, \dots, X_n\} \leq t).$$

Since X_i are iid, $P(X_{\max} \leq t) = P(X_1 \leq t, X_2 \leq t, \dots, X_n \leq t) = \prod_{i=1}^n P(X_i \leq t)$. The cdf of $X_i \sim \text{Unif}(0, \theta)$ is $F_{X_i}(x) = x/\theta$ for $0 \leq x \leq \theta$, 0 for $x < 0$, and 1 for $x > \theta$. So, for $0 \leq t \leq \theta$, $F_{X_{\max}}(t) = (t/\theta)^n$. The pdf of $X_{\max}$ is

$$p_{X_{\max}}(t) = \frac{d}{dt} F_{X_{\max}}(t) = \begin{cases} n \frac{t^{n-1}}{\theta^n} & \text{if } 0 \leq t \leq \theta, \\ 0 & \text{otherwise.} \end{cases}$$

Now, we compute the expectation via integration

$$\begin{aligned}
 \mathbb{E}[X_{\max}] &= \int_0^\theta t \cdot p_{X_{\max}}(t) dt \\
 &= \int_0^\theta t \cdot n \frac{t^{n-1}}{\theta^n} dt \\
 &= \frac{n}{\theta^n} \int_0^\theta t^n dt \\
 &= \frac{n}{\theta^n} \left[\frac{t^{n+1}}{n+1} \right]_0^\theta \\
 &= \frac{n}{\theta^n} \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1} \theta.
 \end{aligned}$$

The bias is $\text{Bias}[\hat{\theta}_{\text{mle}}] = \mathbb{E}[\hat{\theta}_{\text{mle}}] - \theta = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}$. Since the bias is not zero (for finite n), $\hat{\theta}_{\text{mle}} = X_{\max}$ is a biased estimator for θ . It is “asymptotically unbiased” since $\text{bias}[\hat{\theta}_{\text{mle}}] \rightarrow 0$ as $n \rightarrow \infty$.

Exercise 2 (mle for exponential distribution). Let $X_1, \dots, X_n$ be iid RRVs from an exponential distribution for $n \in \mathbb{N}_{>1}$ with unknown rate parameter $\lambda > 0$, i.e., $X \sim (\text{Exp}(\lambda))_{\lambda \in \mathbb{R}_{>0}}$.

- (i) Write down the likelihood function $L(\lambda \mid x)$ and the log-likelihood function $\ell(\lambda \mid x)$ for an observed sample $x = (x_1, \dots, x_n)$ when given only the pdf of an $\text{Exp}(\lambda)$ distribution.
- (ii) Find the maximum likelihood estimator $\hat{\lambda}_{\text{mle}}$ for λ .
- (iii) Calculate the expected value $\mathbb{E}[\hat{\lambda}_{\text{mle}}]$. Is $\hat{\lambda}_{\text{mle}}$ an unbiased estimator for λ ?
[Hint: Since $X_i \sim \text{Exp}(\lambda) = \Gamma(1, \lambda)$, you can use that $S_n = \sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$. Recall that $1/\bar{X}_n = n/S_n$ and use that for $Y \sim \Gamma(\alpha, \beta)$ we have that $\mathbb{E}[1/Y] = \beta/(\alpha - 1)$ if $\alpha > 1$.]

Solution:

- (i) The likelihood function for an observed sample $x = (x_1, \dots, x_n)$ is (using iid)

$$L(\lambda \mid x) = \prod_{i=1}^n p(x_i \mid \lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^n x_i\right),$$

assuming all $x_i \geq 0$. The log-likelihood function is

$$\ell(\lambda \mid x) = \ln(L(\lambda \mid x)) = n \ln \lambda - \lambda \sum_{i=1}^n x_i.$$

- (ii) To find the mle, we differentiate $\ell(\lambda \mid x)$ with respect to λ and set it to zero

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i \stackrel{!}{=} 0.$$

Solving for λ gives

$$\frac{n}{\lambda} = \sum_{i=1}^n x_i \implies \lambda = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}_n}.$$

To confirm it's a maximum, the second derivative is $\frac{d^2\ell}{d\lambda^2} = -\frac{n}{\lambda^2} < 0$ for $\lambda > 0$, so the mle is $\hat{\lambda}_{\text{mle}} = \frac{1}{\bar{X}_n}$.

- (iii) We want to calculate $\mathbb{E}[\hat{\lambda}_{\text{mle}}] = \mathbb{E}[1/\bar{X}_n] = \mathbb{E}[n/S_n]$, where $S_n = \sum_{i=1}^n X_i$. Since $X_i \sim \text{Exp}(\lambda)$, they are also $X_i \sim \Gamma(1, \lambda)$. The sum of n iid $\Gamma(1, \lambda)$ variables is $S_n \sim \Gamma(n, \lambda)$. For a random variable $Y \sim \Gamma(\alpha, \beta)$, if $\alpha > 1$, then $\mathbb{E}[1/Y] = \beta/(\alpha - 1)$. In our case, $S_n \sim \Gamma(n, \lambda)$, so $\alpha = n$ and $\beta = \lambda$. If $n > 1$, $\mathbb{E}[1/S_n] = \lambda/(n - 1)$. Therefore,

$$\mathbb{E}[\hat{\lambda}_{\text{mle}}] = \mathbb{E}[n/S_n] = n \mathbb{E}[1/S_n] = n \frac{\lambda}{n-1} = \frac{n}{n-1} \lambda.$$

The bias is $\text{bias}[\hat{\lambda}_{\text{mle}}] = \mathbb{E}[\hat{\lambda}_{\text{mle}}] - \lambda = \frac{n\lambda}{n-1} - \lambda = \lambda \left(\frac{n}{n-1} - 1 \right) = \frac{\lambda}{n-1}$. Thus, $\hat{\lambda}_{\text{mle}}$ is biased for finite $n > 1$. It is consistent as $\lim_{n \rightarrow \infty} \frac{n}{n-1} \lambda = \lambda$.

Note: For $n = 1$, $\mathbb{E}[1/X_1]$ for $X_1 \sim \text{Exp}(\lambda)$ is $\mathbb{E}[1/X_1] = \int_0^\infty (1/x) \lambda e^{-\lambda x} dx$, which diverges. So $\mathbb{E}[\hat{\lambda}_{\text{mle}}]$ is undefined for $n = 1$.

Exercise 3. Let X be a RVRV with pdf

$$p_X(x) = \begin{cases} 2c^2/x^3 & \text{if } x \geq c, \\ 0 & \text{otherwise,} \end{cases}$$

with $c > 0$.

- (i) Show that p_X is a valid pdf for all $c > 0$.
- (ii) Determine the cdf of X .
- (iii) Let $X_1, \dots, X_n$ be iid RVRVs from this distribution. Find the mle for c .
[Hint: For the likelihood function, consider the case where a sample value is less than c .]

Solution:

- (i) For p_X to be a valid probability density function, it must satisfy both $p_X(x) \geq 0$ for all $x \in \mathbb{R}$ and $\int_{-\infty}^\infty p_X(x) dx = 1$. The first condition is satisfied for $c > 0$, because $2c^2/x^3 > 0$ for all $x \geq c > 0$. Furthermore, for $c > 0$ we have:

$$\begin{aligned} \int_{-\infty}^\infty p_X(x) dx &= \int_c^\infty \frac{2c^2}{x^3} dx = 2c^2 \cdot \lim_{a \rightarrow \infty} \int_c^a x^{-3} dx \\ &= 2c^2 \cdot \lim_{a \rightarrow \infty} \left[-\frac{1}{2} x^{-2} \right]_c^a \\ &= 2c^2 \cdot \left(0 - \left(-\frac{1}{2} c^{-2} \right) \right) = 2c^2 \cdot \left(\frac{1}{2c^2} \right) = 1. \end{aligned}$$

- (ii) The cumulative distribution function of X is given by $F_X(x) = \int_{-\infty}^x p_X(t) dt$. If $x < c$, then $F_X(x) = 0$. For $x \geq c$, we have:

$$F_X(x) = \int_c^x \frac{2c^2}{t^3} dt = 2c^2 \cdot \left[-\frac{1}{2} t^{-2} \right]_c^x = 2c^2 \cdot \left(-\frac{1}{2x^2} - \left(-\frac{1}{2c^2} \right) \right) = 1 - \frac{c^2}{x^2}.$$

Thus, the cdf of X is

$$F_X(x) = \begin{cases} 1 - c^2/x^2 & \text{if } x \geq c, \\ 0 & \text{otherwise.} \end{cases}$$

- (iii) Let $x = (x_1, \dots, x_n)$ be the observed sample values. Since the samples are independent, the likelihood function is given by

$$L(c | x) = \prod_{i=1}^n p_X(x_i) = \prod_{i=1}^n \frac{2c^2}{x_i^3} \chi_{[c, \infty)}(x_i).$$

Let $x_{\min} = \min\{x_1, \dots, x_n\}$. The term $\prod_{i=1}^n \chi_{[c, \infty)}(x_i)$ is 1 if and only if $x_i \geq c$ for all i , which is equivalent to $x_{\min} \geq c$. If $x_{\min} < c$, at least one $p_X(x_i)$ is zero, and thus $L(c | x) = 0$. Thus, the likelihood function is given by

$$L(c | x) = \begin{cases} (2c^2)^n \cdot \prod_{i=1}^n x_i^{-3} & \text{if } c \leq x_{\min}, \\ 0 & \text{otherwise.} \end{cases}$$

For a fixed sample x , the term $\prod_{i=1}^n x_i^{-3}$ is a positive constant. The function $(2c^2)^n$ is strictly increasing in c for $c > 0$. To maximize $L(c | x)$, we must choose c to be as large as possible. However, the likelihood is zero if $c > x_{\min}$. Therefore, the largest possible value for c that does not make the likelihood zero is $c = x_{\min}$. For this reason, $L(c | x)$ is maximized for $c = x_{\min}$. Thus, the mle for c is

$$\hat{c}_{\text{mle}} = \min\{X_1, \dots, X_n\}.$$

Exercise 4. You're throwing $n \in \mathbb{N}$ darts at a target. Let $(X_i)_{i \in [n]}$ be iid random variables with unknown mean μ and unknown standard deviation σ , representing the distance of the i -th arrow from the center of the target. Furthermore, let $\hat{\mu} = \sum_{i=1}^n \lambda_i X_i$ be a convex combination of the individual X_i (i.e., $\lambda_i \geq 0$ for all $i \in [n]$ and $\sum_{i=1}^n \lambda_i = 1$).

(i) Show that $\hat{\mu}$ is an unbiased estimator for μ .

(ii) Prove that $\frac{1}{n} \sum_{i=1}^n \lambda_i^2 \geq \frac{1}{n^2}$.

[Hint: You can do this quickly using Jensen's inequality for an appropriate discrete random variable.]

(iii) Determine the coefficients λ_i that minimize the mse of the estimator $\hat{\mu}$.

Solution:

(i) With the linearity of expectation, the iid assumption, and conditions on λ_i s we compute

$$\mathbb{E}[\hat{\mu}] = \mathbb{E}\left[\sum_{i=1}^n \lambda_i X_i\right] = \sum_{i=1}^n \lambda_i \mathbb{E}[X_i] = \sum_{i=1}^n \lambda_i \mu = \mu \sum_{i=1}^n \lambda_i = \mu.$$

Hence, $\hat{\mu}$ is an unbiased estimator for μ .

(ii) Consider a discrete random variable Z that takes the values $\{\lambda_i\}_{i \in [n]}$ each with equal probability $1/n$. Since $x \mapsto x^2$ is a convex function, we have

$$\frac{1}{n} \sum_{i=1}^n \lambda_i^2 = \sum_{i=1}^n \frac{1}{n} \lambda_i^2 = \mathbb{E}[Z^2] \geq (\mathbb{E}[Z])^2 = \left(\frac{1}{n} \sum_{i=1}^n \lambda_i\right)^2 = \frac{1}{n^2}.$$

(iii) Since $\hat{\mu}$ is an unbiased estimator from part (i), its mse is just its variance $\text{mse}[\hat{\mu}] = \text{var}[\hat{\mu}]$ and we compute

$$\text{var}[\hat{\mu}] = \text{var}\left[\sum_{i=1}^n \lambda_i X_i\right] = \sum_{i=1}^n \lambda_i^2 \text{var}[X_i] = \sum_{i=1}^n \lambda_i^2 \sigma^2 = \sigma^2 \sum_{i=1}^n \lambda_i^2 \geq \frac{\sigma^2}{n},$$

where we use the inequality from (ii). One way to find optimal λ_i s is to find a setting in which the bound is achieved. For $\lambda_i = 1/n$ for each i , we find

$$\text{var}[\hat{\mu}] = \sigma^2 \sum_{i=1}^n \left(\frac{1}{n}\right)^2 = \sigma^2 \cdot n \cdot \frac{1}{n^2} = \frac{\sigma^2}{n},$$

which meets the lower bound with equality. Hence, weighing each sample equally (arithmetic mean) gives the best mse.

Exercise 5. You have a box with $n \in \mathbb{N}_{\geq 1}$ toasters among which a fixed number $d \in \{0, \dots, n\}$ is broken. You draw two toasters uniformly at random from the box.

- (i) What is the probability distribution P_d (depending on d) of the number of broken toasters among the two selected ones?

[Hint: Don't overcomplicate it, this is going to be a distribution over a set with just 3 elements. Feel free to specify the distribution via its pmf.]

- (ii) Assume d to be unknown and we want to estimate it from a (uniform) sample of two toasters. Write down an adequate statistical model and briefly explain it.

- (iii) Provide an unbiased estimator for d in this model and justify your choice.

Solution:

- (i) There can be 0, 1 or 2 broken toasters among the chosen ones, so $\Omega = \{0, 1, 2\}$ and $\mathcal{A} = \mathcal{P}(\Omega)$. Each pair of toasters among the n is equally likely. We draw without replacement and unordered draws, so there are $\binom{n}{2} = n(n-1)/2$ possible pairs. We can consider toasters to be labeled and assume the broken ones to have labels $1, \dots, d$. Then the probability of having zero defective ones in the sampled pair is

$$p_d(0) = \frac{|\{(i, j) \in [n]^2 \mid i, j > d\}|}{\binom{n}{2}} = \frac{\binom{n-d}{2}}{\binom{n}{2}} = \frac{(n-d)(n-d-1)}{n(n-1)}.$$

For both being broken we have

$$p_d(2) = \frac{|\{(i, j) \in [n]^2 \mid i, j \leq d\}|}{\binom{n}{2}} = \frac{\binom{d}{2}}{\binom{n}{2}} = \frac{d(d-1)}{n(n-1)}.$$

And for picking exactly one broken one, we have

$$p_d(1) = 1 - p_d(0) - p_d(2) = \frac{n(n-1)}{n(n-1)} - \frac{(n-d)(n-d-1) + d(d-1)}{n(n-1)} = 2 \frac{nd - d^2}{n(n-1)}.$$

We denote the distribution induced by this pmf via P_d .

- (ii) An adequate statistical model is

$$(\{0, 1, 2\}, \mathcal{P}(\{0, 1, 2\}), (P_d)_{d \in \{0, \dots, n\}})$$

where the number of broken toasters is the unknown parameter $d \in \Theta := \{0, \dots, n\}$. There cannot be fewer than zero or more than n broken toasters. Following (i), there can only be 0, 1 or 2 broken ones in our sample of 2.

- (iii) An intuitive idea is to take the fraction of broken toasters in our sample of two and scale it to all n toasters. So if $\omega \in \Omega$ is the number of defective toasters in our sample, we use $\frac{\omega}{2}n$ as an estimator. Concretely,

$$\hat{d} : \{0, 1, 2\} \rightarrow \mathbb{R}, \omega \mapsto \frac{\omega n}{2}$$

is a $(\mathcal{A}, \mathcal{B})$ -measurable map, thus a statistic and thus a valid estimator for d . To show that it is unbiased we compute

$$\begin{aligned} \mathbb{E}[\hat{d}] &= \sum_{\omega \in \Omega} p_d(\omega) \hat{d}(\omega) \\ &= \frac{n}{2} \left(0 \cdot [\dots] + 1 \cdot \frac{2d(n-d)}{n(n-1)} + 2 \cdot \frac{d(d-1)}{n(n-1)} \right) \\ &= \frac{2d}{2(n-1)} (d-1 + n-d) \\ &= \frac{2d}{2(n-1)} (n-1) = d. \end{aligned}$$

Hence $\mathbb{E}[\hat{d}] = d$ and $\hat{d}$ is an unbiased estimator for d .